Question 9		
(a)	Stage 2: ${}_1^2\mathrm{H} + {}_1^1\mathrm{H} \rightarrow {}_2^3\mathrm{He} + \gamma$ Stage 3: ${}_2^3\mathrm{He} + {}_2^3\mathrm{He} \rightarrow {}_2^4\mathrm{He} + 2{}_1^1\mathrm{H}$ Accept ${}_1^1p$ for proton.	**A1** **A1**
(b)(i)	$E_K = \frac{3}{2}kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times (1.5 \times 10^7 + 273.15) = 3.105 \times 10^{-16} = 3.1 \times 10^{-16}$ J Student need to either show that they convert the temperature to kelvin or explain why they didn't convert to get the marks	**A1**
(ii)	Applying conservation of energy to determine the minimum distance the 2 protons can approach each other: Loss in kinetic energies of 2 protons = Gain in electric potential energy of protons $2E_K - 0 = \frac{e^2}{4\pi\varepsilon_0 x}$ $x = \frac{e^2}{8\pi\varepsilon_0 E_K}$ $= \frac{\left(1.60 \times 10^{-19}\right)^2}{8\pi\varepsilon_0 \left(3.1 \times 10^{-16}\right)}$ $= 3.7 \times 10^{-13}$ m Since d is larger than 10^{-14} m, the protons are not close enough to fuse.	**C1** **A1** **A1**
(c)(i)	The wave function represents the quantum state of a particle, providing all its information such as its position, momentum, etc.	**A1**
(c)(ii)1.	Since the amplitude of wave function after the barrier is non-zero, the probability density function, given by the square of the wave function, also non-zero, implies there is a possibility of finding the particle in this region.	**B1** **B1**
(c)(ii)2.	The wavelength of the wavefunction remains the same, from de Broglie's relationship, the momentum and hence energy of the particle remains the same after passing through the barrier.	**A1** **B1**

(d)(i)	Along the horizontal axis: Magnetic force provides the centripetal force $Bqv_x = mr\omega^2$ $Bqv_x = mv_x\omega$ $\frac{2\pi}{T} = \frac{Be}{m}$ $T = \frac{2\pi \times 9.11\times10^{-31}}{40\times10^{-6}\times1.60\times10^{-19}}$ $= 8.94\times10^{-7}$ s Along the vertical axis: $v_y = 5.0\times10^5 \cos 25^o$ $d = v_y T = (5.0\times10^5 \cos 25^o)\times 8.94\times10^{-7} = 0.41$ m	**B1** **B1** **B1** **A1**
(d)(ii)	Change: Opposite direction. Explain: Proton is positive. Change : Radius will increase. Explain: Proton is more massive. Any other reasonable answer.	**B1** **B1** **B1** **B1**
(d)(iii)	Charged particles in the solar transfer energy to the atoms (or electrons in the atoms) during collisions, exciting them to higher energy levels. When excited atoms/electrons in atom de-excite, photons are emitted.	**A1** **A1**
(e)(i)	Intensity $= \frac{P_{sun}}{4\pi R^2} = \sigma T^4$ $T = \left(\frac{4.0\times10^{26}}{4\pi\times(1.4\times10^{10})^2\times5.7\times10^{-8}}\right)^{1/4}$ $= 1299 = 1300$ K	**B1** **C1** **A1**
(e)(ii)	Help forecast of solar flares, changes in solar wind as modern world depend on devices are sensitive to electromagnetic variations (eg navigation systems, high voltage power grid)	**A1**